

Revised Load Interconnection System Impact Study Report

For Queue Project Q21-1669

State Street 230 kV
January 20, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1669. The Interconnection Customer proposes a 400.4 MW large load (data center) facility interconnecting at the State Street 230 kV substation in the Indiana area. The request was received on February 15, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 608 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,265,000.

3. Network Impacts

3.1 Load Deliverability Analysis

The AEP PETTIT AVENUE - HOHENWALD 100kV is overloaded to 181.8% of its 162 MVA emergency rating for the loss of the WAYNEDALE - STATE STREET SUBSTATION 230kV. The project contributes 8.6% DFAX (34.4 MW) to this overload.

INDUSTRIAL PARK - CORNUTA SUBSTATION 100kV L: 148.0% loading (115 MVA) following the outage of the WAYNEDALE - STATE ST 230kV Line; DFAX = 5.1%.

For the loss of STATE STREET - WAYNEDALE SUBSTATION 230kV, the PEAVINE - HOHENWALD 100kV L loads to 131.4% of its 262 MVA rating. Project DFAX: 12.9%.

The PAGAT SUBSTATION - TRIER 100kV is overloaded to 130.6% of its 126 MVA emergency rating for the loss of the WAYNEDALE - STATE ST 230kV. The project contributes 7.0% DFAX (27.9 MW) to this overload.

SPY RUN - INDUSTRIAL PARK 100kV: 130.0% loading (223 MVA) following the outage of the WAYNEDALE - STATE ST 230kV; DFAX = 8.5%.

SEGS 5 - AEP PETTIT AVENUE 100kV line: 113.1% loading (334 MVA) following the outage of the STATE ST - WAYNEDALE SUBSTATION 230kV; DFAX = 10.1%.

For the loss of STATE ST - WAYNEDALE 230kV, the WAYNEDALE - AEP PETTIT AVENUE 100kV Ckt 1 loads to 103.7% of its 396 MVA rating. Project DFAX: 9.2%.

WAYNEDALE - AEP PETTIT AVENUE 100kV: 103.7% loading (396 MVA) following the outage of the WAYNEDALE - STATE STREET 230kV; DFAX = 9.2%.

WAYNEDALE 230 kV/100 kV XFMR #1: 103.2% loading (268 MVA) following the outage of the STATE STREET - WAYNEDALE 230kV; DFAX = 6.1%.

WAYNEDALE 230/100 kV Auto T2: 103.2% loading (268 MVA) following the outage of the WAYNEALE - STATE ST SUBSTATION 230kV L; DFAX = 6.1%.

For the loss of WAYNEDALE - STATE ST 230kV, the WAYNEDALE 230-100 kV Transformer loads to 103.2% of its 268 MVA rating. Project DFAX: 6.1%.

For the loss of STATE ST - WAYNEDEALE 230kV, the WAYNEDEALE 230/100 kV Autotransformer T4 loads to 103.2% of its 268 MVA rating. Project DFAX: 6.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
AEP PETTIT AVENUE - HOHENWALD 100kV	Rebuild line with higher-capacity conductor	\$2,013,000	56% (\$1,126,000)
CORNUTA - INDUSTRIAL PARK 100kV	Rebuild line with higher-capacity conductor	\$1,013,000	100% (\$1,013,000)
PEAVINE - HHENWALD 100kV L	Rebuild line with higher-capacity conductor	\$1,602,000	24% (\$383,000)
PAGAT - TRIER 100kV	Rebuild line with higher-capacity conductor	\$2,833,000	100% (\$2,833,000)
SPY RUN - INDUSTRIAL PARK 100kV	Rebuild line with higher-capacity conductor	\$1,785,000	22% (\$391,000)
SEGS 5 - AEP PETTIT AVENUE 100kV line	Rebuild line with higher-capacity conductor	\$3,103,000	20% (\$632,000)
AEP PETTIT AVENUE - WAYNEDEALE SUBSTATION 100kV	Reconductor line	\$466,000	18% (\$85,000)
AEP PETTIT AVENUE - WAYNEDEALE 100kV Circuit 2	Reconductor line	\$461,000	18% (\$84,000)
WYNEDALE 230 kV/100 kV XFMR #1	Replace transformer with higher-rated unit	\$7,838,000	64% (\$5,029,000)
WAYNEDEALE 230 kV/100 kV XFMR T2	Replace transformer with higher-rated unit	\$8,555,000	64% (\$5,489,000)
WAYNEDEALE 230-100 kV Transformer	Replace transformer with higher-rated unit	\$7,669,000	64% (\$4,921,000)
WAYNEDEALE 230-100 kV Transformer Bank	Replace transformer with higher-rated unit	\$6,669,000	64% (\$4,279,000)
	Total allocated to project		\$26,265,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.